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MYSQL *project*

PROJECT DETAIL

SQL + Business Insights Project

This project analyzes an online bookstore dataset using MySQL, consisting of Books, Customers, and Orders tables. The goal is to extract meaningful business insights related to sales performance, customer behavior, and inventory management.

Using SQL concepts such as JOINs, GROUP BY, aggregate functions, and filtering, the project answers key business questions like identifying top-selling books, high-value customers, revenue trends, and stock availability.

The analysis helps in understanding sales patterns, customer preferences, and inventory status, demonstrating how SQL can be effectively used for data-driven decision-making in a business environment.

```
use business_start;          {----DATASET

select * from books;         {---TABLE 1
select * from customer;      {---TABLE 2
select * from orders;        {---TABLE 3
```

-- BASIC QUESTIONS --

-- 1) Retrieve all books in the "Fiction" genre

```
select * from books
where genre = "fiction"
order by stock;
```

```
-- 2) Find books published
after the year 1950

select * from books
where publish_year > 1950
order by publish_year;
```

-- 3) List all customers from the Canada

```
select * from customer
where country = "canada";
```

-- 4) Show orders placed in November 2023

```
select * from orders
where order_date between '2023-11-1'
and '2023-11-30'
order by order_date;
```

-- 5) Retrieve the total stock of books available

```
select sum(stock) from books;
```

-- 6) Find the details of the most expensive book

```
select * from books
order by price desc
limit 1;
```

-- 7) Show all customers who ordered more than 1 quantity of a book

```
select * from orders
where quantity>1
order by quantity;
```

```
-- 8) Retrieve all orders where
the total amount exceeds $20

select * from orders
where total_amount>20;
```

-- 9) List all genres available in the Books table

```
select genre from books;
```

-- 10) Find the book with the lowest stock

```
select * from books
order by stock;
```

-- 11) Calculate the total revenue generated from all orders

```
select sum(total_amount) as total_revenue from orders;
```

-- ADVANCE QUESTIONS --

-- 1) Retrieve the total number of books sold for each genre

```
select b.genre , sum(o.quantity) as total_no_booksold
from orders as o
join books as b
on b.book_id = o.book_id
group by genre;
```

-- 2) Find the average price of books in the "Fantasy" genre

```
select genre, avg(price) from books
where genre = "fantasy";
```

-- 3) List customers who have placed at least 2 orders

```
select customer_id,count(quantity) from orders
group by customer_id
having count(quantity) >1;
```

-- 4) Find the most frequently ordered book

```
select o.book_id ,b.title, count(o.order_id)
from orders o
join books b on o.book_id = b.book_id
group by o.book_id , b.title
order by count(order_id) desc
limit 1;
```

-- 5) Show the top 3 most expensive books of 'Fantasy' Genre

```
select * from books
where genre = "fantasy"
order by price desc
limit 3;
```

-- 6) Retrieve the total quantity of books sold by each author

```
select b.author , sum(o.quantity) from orders as o
join books b on b.book_id = o.book_id
group by b.author;
```

-- 7) List the cities where customers who spent over $30 are located

```
select distinct(c.city)  from orders as o
join customer c on c.customer =
o.customer_id
where total_amount>30;
```

-- 8) Find the customer who spent the most on orders

```
select c.customer , c.name, sum(o.total_amount) from
orders as o
join customer c on c.customer = o.customer_id
group by c.customer ,c.name
order by sum(o.total_amount) desc
limit 1;
```

-- 9) Calculate the stock remaining after fulfilling all orders

```
select b.book_id,b.title,b.stock, coalesce(sum(o.quantity),0)
as order_quantity,
b.stock- coalesce(sum(o.quantity),0) as remainning_quantity
from books as b
left join orders as o
on b.book_id = o.book_id
group by b.book_id
order by b.book_id;
```